MIL-STD-9A
26 MAY 1960
SUPERSEDING
MIL-STD-9
15 JUNE 1955

MILITARY STANDARD

SCREW THREAD CONVENTIONS AND METHODS OF SPECIFYING

UNITED STATES
GOVERNMENT PRINTING OFFICE
WASHINGTON : 1960

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DEPARTMENT OF DEFENSE
ARMED FORCES SUPPLY SUPPORT CENTER
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SCREW THREAD CONVENTIONS
AND METHODS OF SPECIFYING

MIL-STD-9A

1. This standard has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force, effective 26 May 1960.

2. In accordance with established procedure, the Ordnance Corps, Bureau of Naval Weapons, and the Air Force have been designated as the Army-Navy-Air Force custodians of this standard.

3. Provisions of this standard are the subject of international standardization agreement. When revision or cancellation of this standard is proposed, the departmental custodians will inform their respective Departmental Standardization Offices so that appropriate action may be taken respecting the international agreement concerned.

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MILITARY STANDARD
SCREW THREAD CONVENTIONS AND METHODS OF SPECIFYING

1. GENERAL

1.1 Purpose. The purpose of this standard is to establish uniform practices governing the representation and designation of screw threads on drawings prepared by or for the Department of Defense.

1.2 Scope. This standard has application to screw threads covered by Handbook H28, Screw Thread Standards for Federal Services. The scope of this standard includes: (a), Definitions; (b), Representations; (c), Designations; (d), Specifications; (e), Dimensioning; and (f), Recommended Selection.

2. REFERENCED DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids form a part of this standard:

STANDARDS

MIL–STD–10 — Surface Roughness, Waviness, and Lay

MS–33537 — Insert, Standard Dimensions for Coarse and Fine Thread Helical Coil, Assembly.

SPECIFICATIONS

Federal Specification ZZ–H–466, Hose, Gasoline, Rubber — Metal.

(Copies of the above listed standards and specifications may be obtained in the manner stated on the last page of this standard).

OTHER PUBLICATIONS

U. S. Department of Commerce, National Bureau of Standards, Handbook H28 "Screw-Thread Standards for Federal Services".

(Copies may be obtained from the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.)

3. DEFINITIONS

3.1 Screw thread terms. For purposes of this standard, the definitions herein apply. Other definitions are given throughout the text, as needed, and also in Handbook H28.

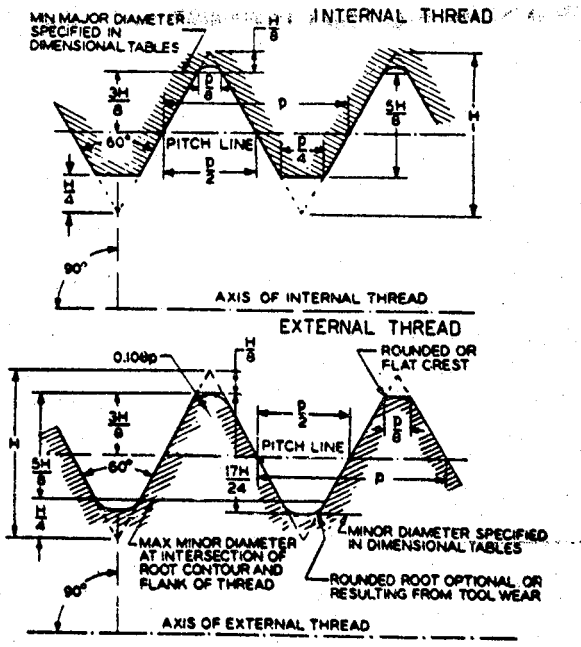


FIGURE 1. *Design forms at maximum material condition (MMC)*

3.2 Unified form thread. The unified form thread conforms to the proportions of profile shown in Figure 1, the limits of size for which are obtained from the Unified tabulations or formulations prescribed by Handbook H28. These threads are designated: UN, UNC, UNF, UNEF, UNS, or UNM.

3.3 American National form thread. The American National form thread has substantially the same proportions of profile as corresponding features of the Unified form thread. The principal differences between the Unified and the American National form

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threads are in the application of allowances, differences in amount of pitch diameter tolerance applied to internal and external threads, and the variation of tolerance with size. These threads are designated: N, NC, NF, NEF, or NS.

3.4 **Thread series.** A thread series is a group of diameter-pitch combinations, different series being distinguished from each other by the number of threads per inch applied to a specific diameter.

3.5 **Standard series threads.** Standard series threads are threads of Unified form, having diameter-pitch combinations listed in Table I; the limits of size of which conform in all detail to the tabulations for Standard Series threads provided in Handbook H28 (1957) Section III. Dimensional limits, allowances and tolerances for UNC, UNF, 4UN, 6UN and 8UN are based on a length of engagement equal to the basic major (nominal) diameter and are applicable to lengths of engagement up to 1½ diameters. UNEF, 12UN, 16UN, 20UN, 28UN and 32UN are based on a length of engagement of 9 pitches and are applicable to lengths of engagement from 5 to 15 pitches.

3.6 **Special series (selected combinations) threads.** Special series threads are threads of Unified form, symbolized "UNS", having the selected combinations listed in Table II; the limits of size of which conform in all details to those computed from the Unified tabulations or formulations provided in Handbook H28 (1957) Section IV. Dimensional limits, allowances and tolerances are based on a length of engagement of nine times pitch (up to 15 x p).

3.7 **Special threads.** Special threads, as defined herein, are threads having features (such as profile, diameter-pitch combinations, limits of size, or lengths of engagement) which differ from those applicable to corresponding diameter-pitch combinations as listed in Tables I or II. Special threads which conform in all details to Unified tabulations or formulations are symbolized "UNS." Various degrees of specialization in the design of special threads are set forth in Handbook H28 (1957) Section IV.

3.8 **Modified threads.** Modified threads are standard series, special series, or special threads which conform in all detail to the Unified or American National tabulations or formulations, with the single exception that the limits of size of the major diameter of the external thread have been reduced, or the limits of size of the minor diameter of the internal thread have been increased from the dimensional limits derived for such threads by the tabulations or formulations prescribed by Handbook H28, but without changing the class of thread or pitch diameter limits.

3.9 **Unified miniature screw threads.** Unified miniature screw threads are threads of Unified form, covering a diameter range from .0118 to .0051 inches, and thus supplementing the standard series threads which begin at No. 0 (.0600). Established standards for Unified miniature screw threads are presented by Handbook H28 (1957) Section V.

3.10 **Special form threads.** Threads are considered to be of special form when they are other than standard profile. This includes change from standard included angle, nonsymmetrical form, variation from standard flank length for an otherwise standard thread form except as covered by modified thread examples (see 5.2.4.5, 5.2.4.6) or elliptical or radius thread forms.

3.11 **American Standard taper and straight pipe thread (except Dryseal and hose coupling types).** The American Standard taper pipe thread is a form of screw thread used on pipe and pipe fittings and it is characterized by a fine pitch and a taper of 1 in 16 (.75 in/ft) on the diameter. The straight pipe thread conforms with the American Standard taper pipe thread with respect to pitch and depth of thread. Standards for taper and straight pipe threads are described in Handbook H28 (1957) Section VII.

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TABLE I. *Standard series — unified screw threads*

Sizes		Basic major diameter	Threads per inch											Sizes
			Series with graded pitches			Series with constant pitches								
Primary	Secondary		Coarse UNC	Fine UNF	Extra fine UNEF	4UN	6UN	8UN	12UN	16UN	20UN	28UN	32UN	
0		0.0600	---	80	---	---	---	---	---	---	---	---	---	0
	1	0.0730	64	72	---	---	---	---	---	---	---	---	---	1
2		0.0860	56	64	---	---	---	---	---	---	---	---	---	2
	3	0.0990	48	56	---	---	---	---	---	---	---	---	---	3
4		0.1120	40	48	---	---	---	---	---	---	---	---	---	4
5		0.1250	40	44	---	---	---	---	---	---	---	---	---	5
6		0.1380	32	40	---	---	---	---	---	---	---	---	UNC	6
8		0.1640	32	36	---	---	---	---	---	---	---	---	UNC	8
10		0.1900	24	32	---	---	---	---	---	---	---	---	UNF	10
	12	0.2160	24	28	32	---	---	---	---	---	---	UNF	UNEF	12
1/4		0.2500	20	28	32	---	---	---	---	---	UNC	UNF	UNEF	1/4
5/16		0.3125	18	24	32	---	---	---	---	---	20	28	UNEF	5/16
3/8		0.3750	16	24	32	---	---	---	---	UNC	20	28	UNEF	3/8
7/16		0.4375	14	20	28	---	---	---	---	16	UNF	UNEF	32	7/16
1/2		0.5000	13	20	28	---	---	---	---	16	UNF	UNEF	32	1/2
9/16		0.5625	12	18	24	---	---	---	UNC	16	20	28	32	9/16
5/8		0.6250	11	18	24	---	---	---	12	16	20	28	32	5/8
	11/16	0.6875	---	---	24	---	---	---	12	16	20	28	32	11/16
3/4		0.7500	10	16	20	---	---	---	12	UNF	UNEF	28	32	3/4
	13/16	0.8125	---	---	20	---	---	---	12	16	UNEF	28	32	13/16
7/8		0.8750	9	14	20	---	---	---	12	16	UNEF	28	32	7/8
	15/16	0.9375	---	---	20	---	---	---	12	16	UNEF	28	32	15/16
1		1.0000	8	12	20	---		UNC	UNF	16	UNEF	28	32	1
	1 1/16	1.0625	---	---	18	---	---	8	12	16	20	28	---	1 1/16
1 1/8		1.1250	7	12	18	---	---	8	UNF	16	20	28	---	1 1/8
	1 3/16	1.1875	---	---	18	---	---	8	12	16	20	28	---	1 3/16
1 1/4		1.2500	7	12	18	---	---	8	UNF	16	20	28	---	1 1/4
	1 5/16	1.3125	---	---	18	---	---	8	12	16	20	28	---	1 5/16
1 3/8		1.3750	6	12	18	---	UNC	8	UNF	16	20	28	---	1 3/8
	1 7/16	1.4375	---	---	18	---	6	8	12	16	20	28	---	1 7/16
1 1/2		1.5000	6	12	18	---	UNC	8	UNF	16	20	28	---	1 1/2
	1 9/16	1.5625	---	---	18	---	6	8	12	16	20	---	---	1 9/16
1 5/8		1.6250	---	---	18	---	6	8	12	16	20	---	---	1 5/8
	1 11/16	1.6875	---	---	18	---	6	8	12	16	20	---	---	1 11/16
1 3/4		1.7500	5	---	---	---	6	8	12	16	20	---	---	1 3/4
	1 13/16	1.8125	---	---	---	---	6	8	12	16	20	---	---	1 13/16

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TABLE I. *Standard series — unified screw threads—Continued*

Sizes		Basic major diameter	Threads per inch											Sizes
			Series with graded pitches			Series with constant pitches								
Primary	Secondary		Coarse UNC	Fine UNF	Extra fine UNEF	4UN	6UN	8UN	12UN	16UN	20UN	28UN	32UN	
1⅞		1.8750	....	....	....	....	6	8	12	16	20	....	....	1⅞
	1 15/16	1.9375	....	....	....	....	6	8	12	16	20	....	....	1 15/16
2		2.0000	4½	....	....	....	6	8	12	16	20	....	....	2
	2⅛	2.1250	....	....	....	....	6	8	12	16	20	....	....	2⅛
2¼		2.2500	4½	....	....	....	6	8	12	16	20	....	....	2¼
	2⅜	2.3750	....	....	....	....	6	8	12	16	20	....	....	2⅜
2½		2.5000	4	....	....	UNC	6	8	12	16	20	....	....	2½
	2⅝	2.6250	....	....	....	4	6	8	12	16	20	....	....	2⅝
2¾		2.7500	4	....	....	UNC	6	8	12	16	20	....	....	2¾
	2⅞	2.8750	....	....	....	4	6	8	12	16	20	....	....	2⅞
3		3.0000	4	....	....	UNC	6	8	12	16	20	....	....	3
	3⅛	3.1250	....	....	....	4	6	8	12	16	....	....	....	3⅛
3¼		3.2500	4	....	....	UNC	6	8	12	16	....	....	....	3¼
	3⅜	3.3750	....	....	....	4	6	8	12	16	....	....	....	3⅜
3½		3.5000	4	....	....	UNC	6	8	12	16	....	....	....	3½
	3⅝	3.6250	....	....	....	4	6	8	12	16	....	....	....	3⅝
3¾		3.7500	4	....	....	UNC	6	8	12	16	....	....	....	3¾
	3⅞	3.8750	....	....	....	4	6	8	12	16	....	....	....	3⅞
4		4.0000	4	....	....	UNC	6	8	12	16	....	....	....	4
	4⅛	4.1250	....	....	....	4	6	8	12	16	....	....	....	4⅛
4¼		4.2500	....	....	....	4	6	8	12	16	....	....	....	4¼
	4⅜	4.3750	....	....	....	4	6	8	12	16	....	....	....	4⅜
4½		4.5000	....	....	....	4	6	8	12	16	....	....	....	4½
	4⅝	4.6250	....	....	....	4	6	8	12	16	....	....	....	4⅝
4¾		4.7500	....	....	....	4	6	8	12	16	....	....	....	4¾
	4⅞	4.8750	....	....	....	4	6	8	12	16	....	....	....	4⅞
5		5.0000	....	....	....	4	6	8	12	16	....	....	....	5
	5⅛	5.1250	....	....	....	4	6	8	12	16	....	....	....	5⅛
5¼		5.2500	....	....	....	4	6	8	12	16	....	....	....	5¼
	5⅜	5.3750	....	....	....	4	6	8	12	16	....	....	....	5⅜
5½		5.5000	....	....	....	4	6	8	12	16	....	....	....	5½
	5⅝	5.6250	....	....	....	4	6	8	12	16	....	....	....	5⅝
5¾		5.7500	....	....	....	4	6	8	12	16	....	....	....	5¾
	5⅞	5.8750	....	....	....	4	6	8	12	16	....	....	....	5⅞
6		6.0000	....	....	....	4	6	8	12	16	....	....	....	6

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TABLE II. *Special series — unified screw threads*
(selected combinations)
thread series symbol UNS, classes 2A and 2B

Nominal size	Threads per inch											
	10	12	14	16	18	24	27	28	36	40	48	56
No. 10								28	36	40	48	56
No. 10												
No. 12									36	40	48	56
No. 12												
¼						24	27		36	40	48	56
¼												
5/16							27		36	40	48	
5/16												
⅜					18		27		36	40		
⅜												
.390							27					
7/16					18	24	27					
7/16												
½		*12	14		18	24	27					
½												
9/16			14				27					
9/16												
5/8			14				27					
5/8												
¾			14		18	24	27					
¾												
⅞	10				18	24	27					
⅞												
1	10		**14		18	24	27					
1												
1⅛	10		14			24						
1⅛												
1¼	10		14			24						
1¼												
1⅜	10		14			24						
1⅜												

Nominal size	Threads per inch					
	10	12	14	16	18	24
1½	10		14			24
1½						
1⅝	10		14			24
1⅝						
1¾	10		14		18	
1¾						
1⅞	10		14		18	
1⅞						
2	10		14		18	
2						
2 1/16				*16		
2 1/16				*16		
2¼	10		14		18	
2¼						
2 5/16				*16		
2 7/16				*16		
2½	10		14		18	
2½						
2¾	10		14		18	
2¾						
3	10		14		18	
3						
3¼	10		14		18	
3¼						
3½	10		14		18	
3½						
3¾	10		14		18	
3¾						
4	10		14			
4						
4¼	10		14			
4¼						
4½	10		14			
4½						
4¾	10		14			
4¾						
5	10		14			
5						
5¼	10		14			
5¼						
5½	10		14			
5½						
5¾	10		14			
5¾						
6	10		14			
6						

* Classes 2A, 2B, 3A, 3B.
** Classes 1A, 1B, 2A, 2B, 3A, 3B.

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3.12 Dryseal American standard pipe thread. The Dryseal pipe thread consists of four types. The external Dryseal pipe thread is tapered, whereas the internal Dryseal pipe thread may be either straight or tapered. The significant feature of the Dryseal thread is controlled truncation of the crest and root to assure metal to metal contact coincident with, or prior to, flank contact. Contact at the crest and root prevents spiral leakage and insures pressure-tight joints without the use of a lubricant or sealer. Standards for Dryseal pipe threads are presented in Handbook H28 (1957) Section VIII.

3.13 Gas cylinder valve outlet and inlet threads. Standards for these threads are presented in Handbook H28 (1957) Section IX.

3.13.1 Gas cylinder valve outlet threads are used on the outlet valves of compressed gas cylinders. Where practicable, right-hand threads are used for non-fuel gases and left-hand threads for fuel gases.

3.13.2 Gas cylinder valve inlet threads are used on the inlet, neck, or valve to cylinder connection of compressed gas cylinders.

3.14 American National hose coupling and fire-hose coupling threads. The American National hose coupling and fire-hose coupling threads are used for threaded parts of hose couplings, valves, nozzles, and other fittings used in direct connection with hose intended for fire protection or for domestic, industrial, and general services. Standards for these threads are presented in Handbook H28 (1957) Section X.

3.15 Acme screw threads. Acme screw threads (29 degree) are used primarily for traversing motion and power transmission. The three classes of general purpose threads have clearances on all diameters for free movement and may be used in assemblies with the internal thread rigidly fixed and movement of the screw in a direction perpendicular to the axis limited by its bearing or bearings. All five classes of centralizing threads have a limited clearance at the major diameters of external and internal threads so that a bearing at the major diameter maintains approximate alignment of the thread axis and prevents wedging on the flanks of the thread. Standards for general purpose and centralizing Acme screw threads are presented in Handbook H28 (1957) Section XII.

3.16 Stub Acme screw threads. Stub Acme threads are also used for traversing and power transmission purposes, but their use is generally confined to those unusual applications where a coarse pitch thread of shallow depth is required due to mechanical or metallurgical considerations. A series of diameters and associated pitches of Stub Acme threads has been selected and are recommended as preferred. There is established only one thread class of Stub Acme threads for general usage. This class corresponds to the Class 2G (General purpose) of the American Standard Acme threads. Standards for Stub Acme threads are presented in Handbook H28 (1957) Section XIII.

3.17 Buttress screw threads. The Buttress form of thread has certain advantages in applications involving exceptionally high stresses in one direction only, along the thread axis. Standards for Buttress threads are presented in Handbook H28 (1957) Section XIV.

4. GENERAL REQUIREMENTS

Not applicable.

5. DETAIL REQUIREMENTS

5.1 Thread representation.

5.1.1 *Methods of representation.* Three methods described herein are in general use for representing "V" type screw threads on drawings: (a), Simplified; (b), Schematic;

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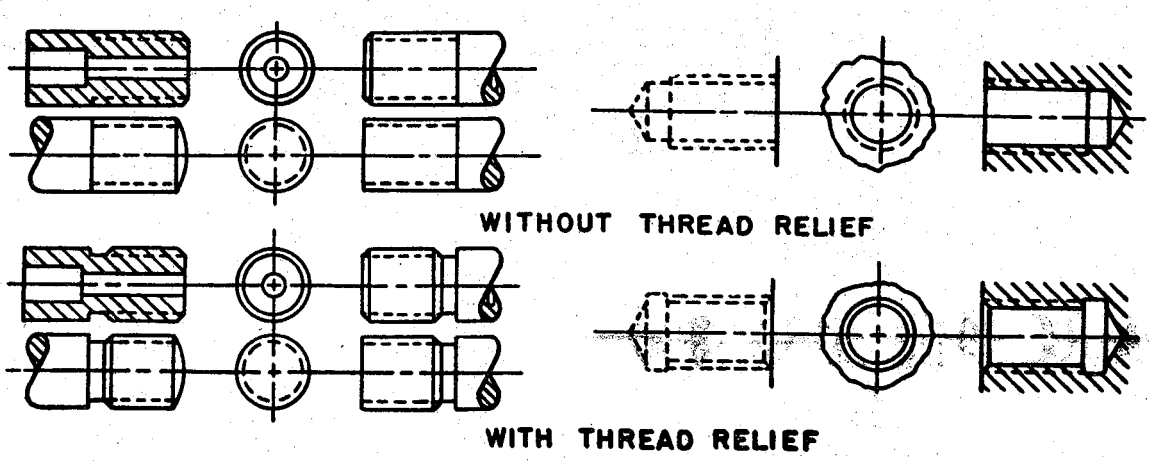


FIGURE 2. *Simplified method of thread representation*

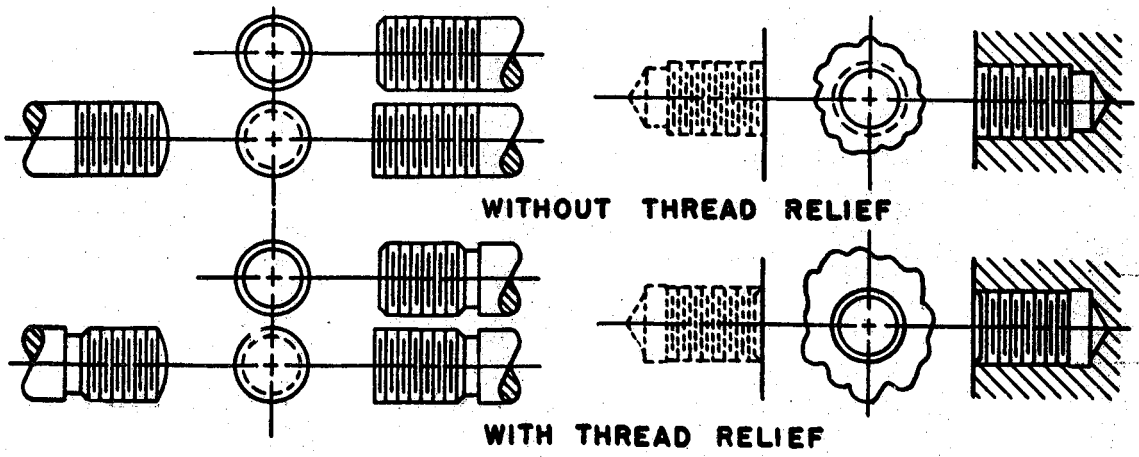


FIGURE 3. *Schematic method of thread representation*

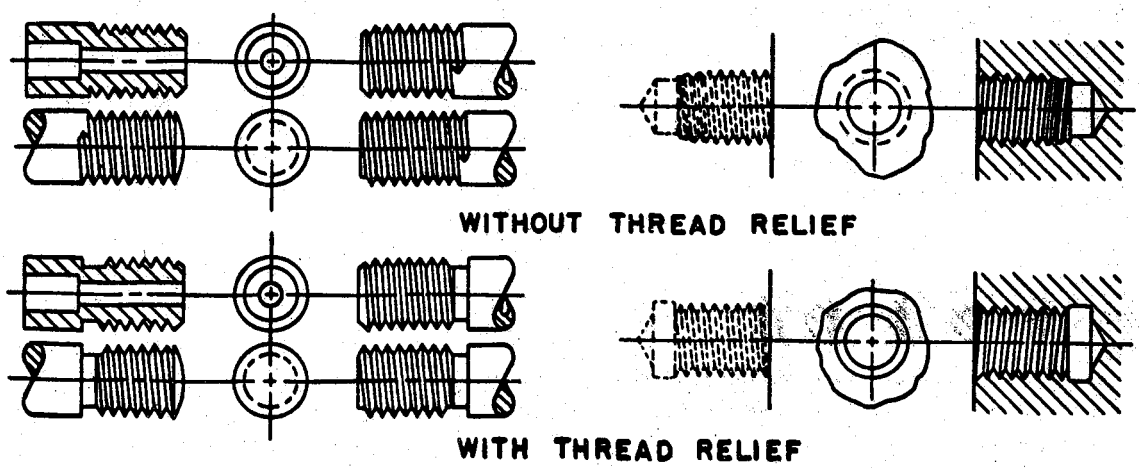


FIGURE 4. *Detailed method of thread representation*

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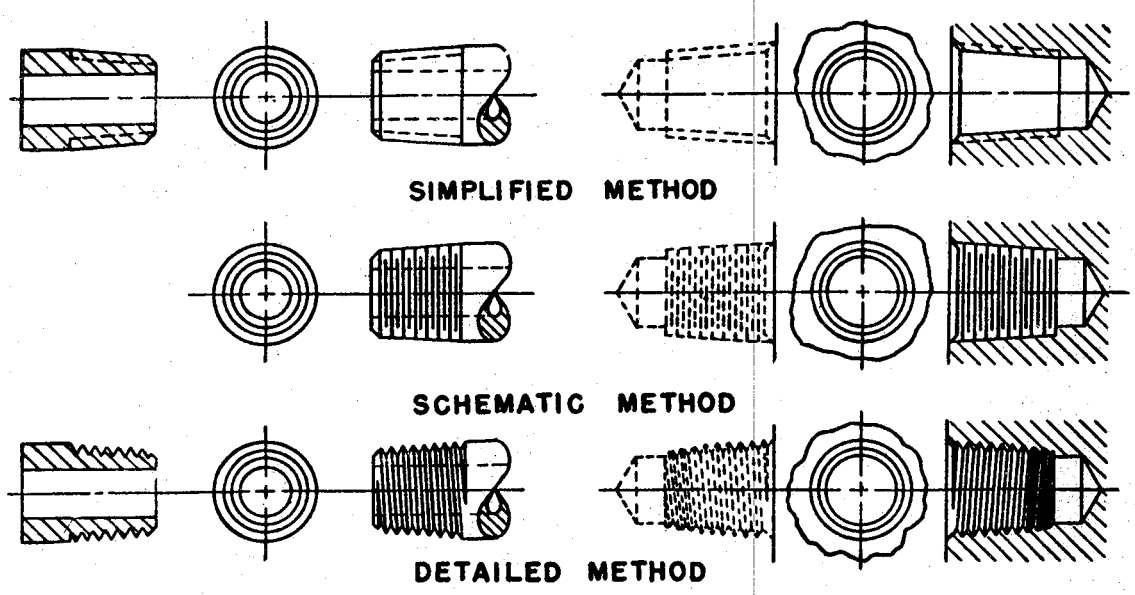


FIGURE 5. *Tapered pipe thread representation*

(c), Detailed. The particular method selected for depicting a thread shall be governed by the amount of detail required on the type of drawing involved. Acme, Stub Acme, and Buttress type threads may be represented by the schematic method except when a pictorial representation of exact thread geometry is essential for end purpose requirements.

5.1.1.1 *Simplified method.* The simplified method, illustrated in Figure 2, is the most easily constructed and should be used wherever practicable. The external thread is represented axially by a solid line for the major diameter and a dash line for the minor diameter. The internal thread, in an axial section, is represented by a solid line for the minor diameter and a dash line for the major diameter. Hidden threads, both external and internal, are represented by dash lines.

5.1.1.2 *Schematic method.* The schematic method, illustrated in Figure 3, employs lines representing the major diameter, the crest and the root of the thread. Axially, the lines representing the crest are drawn to contact those representing the major diameter; the lines representing the root to a length approximating the minor diameter. Preferably, the lines representing the crest and the root are drawn perpendicular to the thread axis. The spacing of the lines representing the crest and the root shall be uniform, but need not indicate the actual pitch or the number of threads required.

5.1.1.3 *Detailed method.* The detailed method, illustrated in Figure 4, may either be a true representation of the thread, or a modified representation of the form, such as sharp V's and straight slanting lines for the V-type threads. Use of the detailed method is rarely necessary, and generally should be restricted to assembly drawings on which the utilization of other methods for thread representation would cause misinterpretation.

5.1.2 *Representation of pipe threads.* Tapered and straight pipe threads are depicted on drawings in the manner described in paragraph 5.1.1, except that tapered pipe threads shall be drawn showing a taper approximately 1 to 8 (¾ in. = 1 ft.). Methods of rep-

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resenting tapered pipe threads are illustrated in Figure 5.

5.1.3 *Representation of thread end views, with or without chamfers and countersinks.* End view representations of threads are identical for the simplified, schematic, and the detailed methods. For an external thread having a chamfered end, two full circles representing the chamfered end, are shown; for a plain, oval or rounded end, a full circle is shown for the major diameter and a dash circle for the minor diameter. For an internal thread with countersink, two full circles are shown; but without countersink, a full circle and a dash circle are shown, representing, respectively, the minor and the major thread diameters. End view representations are illustrated in Figures 2, 3, 4 and 5.

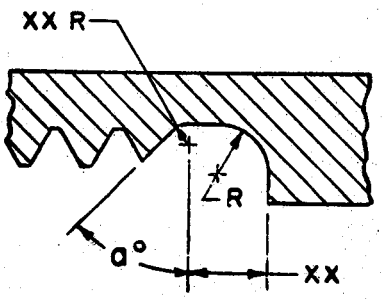


FIGURE 6. *Enlarged thread detail*

5.1.4 *Enlarged thread detail.* In those instances where threads are shown as a portion of a greatly enlarged detail, the representation of exact thread geometry may be justified. For illustration, see Figure 6.

5.1.5 *Combination of methods.* All three methods of thread representation, as described in par. 5.1.1, may be combined on the same drawing, consistent with end purpose requirements. For illustration, see Figure 7.

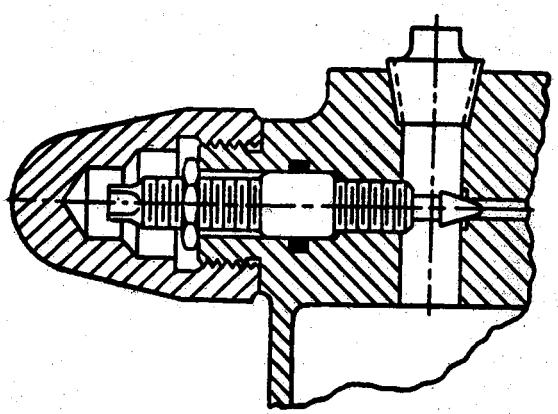


FIGURE 7. *Illustration of combined methods of thread representation*

5.1.6 *Representation of helical coil thread insert.* Helical coil thread inserts are shown by the convention illustrated in Figure 8.

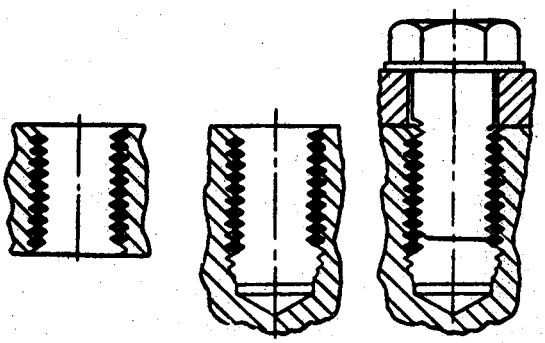
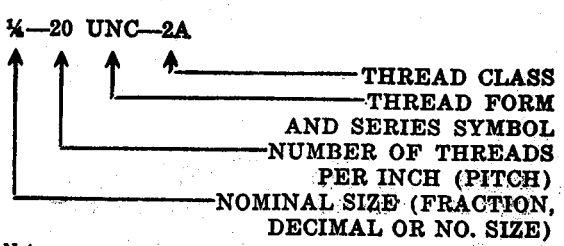


FIGURE 8. *Representation of helical coil thread insert*

5.2 Thread designation.

5.2.1 *Basic designation.* The minimum of information required in the designation of standard series threads consists of the following groups, in sequence, each group being separated by a dash: nominal size — number of threads per inch and thread series symbol — thread class.



Note:
Consistent with ABC agreement made subsequent to the issuance of Handbook H28 (1957) Part I, when the dimensional limits for threads, including those in the numbered series, conform in all detail to Unified tabulations or formulations, the letter "U" shall appear in the thread form and series symbol.

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5.2.1.1 *Nominal size.* Preferably, the nominal size shall be expressed in fractional form, except for the numbered series and special sizes not convertible to commonly used fractional form. Alternatively, decimal equivalents designating the nominal size may be used in lieu of common fractions. When expressed in inches, or fraction thereof, the nominal size shall be so designated without inch marks, e.g., 7/16 or .4375, 1 1/4 or 1.2500, etc. The nominal size for the numbered series threads shall be designated either by the number, e.g., No. 10, or by the decimal equivalent, e.g., .1900. Decimal equivalents of the nominal size, when used, shall have the same number of decimal places as the modifying tolerance.

5.2.1.2 *Number of threads.* This number, preceded by a dash (—), designates the number of single threads per inch (pitch equal to lead). Multiple threads are specified by pitch and lead and also by the number of "starts" affixed to the thread designation. Examples of single, double and triple starts are illustrated in Table IV (examples 1, 12 and 13).

5.2.1.3 *Thread form and series symbol.* This symbol identifies the form of thread in association with the thread series. Typical examples of these symbols are listed in Table III.

5.2.1.4 *Thread class.* Classes of threads are distinguished from each other by the amount

TABLE III. *Thread form and series symbols*

Thread form and series symbol	Description
ACME	Acme thread
STUB ACME	Stub Acme thread
60° STUB	60° Stub thread
NC	*American National coarse thread
NF	*American National fine thread
NEF	*American National extra-fine thread
8N	*American National 8-thread
12N	*American National 12-thread
16N	*American National 16-thread
NS	*Special threads of American National form
N BUTT	National Buttress thread
NGO	American National gas outlet thread
NGS	American National gas straight thread
NGT	American National gas taper thread
SGT	American National special gas taper thread
NH	American National hose coupling and fire-hose coupling thread
NPSH	American National hose coupling thread (iron pipe sizes)
AMO	American Standard microscope objective and nose-piece thread
	American Standard surveying instrument mounting thread
UNC	Unified coarse thread
UNF	Unified fine thread
UNEF	Unified extra-fine thread
UNM	Unified miniature thread
4UN, 6UN, 8UN, 12UN, 16UN, 20UN, 28UN, 32UN	Unified constant-pitch series
UNS	Special threads of Unified form
27UNS	27-thread Unified special thread series for use on thin wall tubing

* Superseded by Unified thread series.

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of tolerance, or the amount of tolerance and allowance. For threads computed to Unified formulation, classes 1A, 2A and 3A apply to external threads and classes 1B, 2B and 3B apply to internal threads. For the American National series, the class of thread (formerly "class of fit") is designated, for external and internal threads, by a number only, preceded by a dash, e.g., —2, —3.

5.2.1.4.1 *Exceptions to thread class designation.* Some threads are not identified by thread class designations. For examples, see Table IV.

5.2.1.5 *Hand.* Designations of screw threads, except as set forth herein, shall be interpreted to mean "RH", unless "LH" is included in the thread call out. For gas cylinder valve outlet threads of NGO series, the designations "RH" or "LH", as applicable, shall be included in the thread call out. For examples, see Table IV (examples 11, 15 and 16).

5.2.2 *Arrangement of basic designations.* Typical examples of arrangements of basic designations are listed in Table IV.

5.2.3 *Supplementary thread data.* The designations for special series, modified, and special threads require additional data depending upon the degree of specialization, as illustrated in the pertinent examples provided herein.

TABLE III. *Thread form and series symbols — Continued*

Thread form and series symbol	Description
	American Standard Pipe Threads (except Dryseal and Hose Coupling Types)
NPT	American Standard taper pipe threads for general use
NPTR	American Standard taper pipe thread for railing joints
NPSC	American Standard straight pipe thread in pipe couplings
NPSL	American Standard straight pipe thread for loose-fitting mechanical joints with locknuts
NPSM	American Standard straight pipe thread for free-fitting mechanical joints for fixtures
ANPT	Aeronautical taper pipe thread
	Dryseal American Standard Pipe Threads
DRYSEAL F–PTF	Dryseal American Standard fine taper pipe thread
DRYSEAL NPSF	Dryseal American Standard fuel internal straight pipe thread
DRYSEAL NPSI	Dryseal American Standard intermediate internal straight pipe thread
DRYSEAL NPTF	Dryseal American Standard taper pipe thread
DRYSEAL PTF–SAE SHORT	Dryseal SAE short taper pipe thread
DRYSEAL PTF–SPL SHORT	Dryseal American Standard special short pipe thread
DRYSEAL PTF–SPL EXTRA SHORT	Dryseal American Standard special extra short pipe thread
DRYSEAL SPL–PTF	Dryseal American Standard special diameter-pitch combination pipe thread

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TABLE IV. *Arrangement of basic designations*

EX	Nominal Size	Dash	Number of threads per inch	Thread Form and Series Symbol	Dash	Thread Class	Dash	Hand	
1.	⅝	—	16	UNC	—	2A			
2.	⅝	—	24	NF	—	2			
3.	1¾	—	4	ACME	—	2G			
4.	.800	—	36	AMO					
5.	1¼	—	11½	NPT					or 1.250–11.5 NPT
6.	⅛	—	27	NPSC					
7.	½	—	14	NPTR					
8.	⅛	—	27	NPSM					
9.	⅛	—	27	NPSL					
10.	¾	—	14	NPSH					
11.	No. 10	—	24	UNC	—	2B	— LH	—	
12.	2½	—	.4p–.8L	ACME	—	3G			(See Note 1)
13.	1½	—	12	UNF	—	2A	— LH	—	3 START (See Note 2)
14.	½	—	20	STUB ACME					
15.	.895	—	18	NGO			— RH	—	EXT (See Note 3)
16.	.899	—	18	NGO			— RH	—	INT (See Note 3)
17.	4	—	8	←(N BUTT	—	2	— LH		(See Note 4)
18.	⅝	—	20	(←N BUTT	—	1 (See Note 5)			
19.	⅛	—	27	DRYSEAL NPTF					
20.	⅛	—	27	DRYSEAL PTF–SAE SHORT					
21.	⅛	—	27	DRYSEAL NPSF					
22.	⅛	—	27	DRYSEAL NPSI					

Note 1 For multiple start threads, except American National or Unified as shown in example 13, both the lead and the pitch shall be specified instead of the number of threads per inch.

Note 2 For Unified or American National multiple start screw threads, the designation is the basic designation plus the number of starts.

Note 3 American National Gas Outlet threads fall into four divisions, right-hand or left-hand (RH or LH) and external or internal (EXT or INT). These four basic divisions are so vital to prevent undesirable cross-connecting that the full designation of each gas outlet thread includes these terms.

Note 4 This indicates a basic major diameter of 4 in., 8 threads per in., internal member or screw to pull, National Buttress form of thread, class 2, left hand.

Note 5 This indicates a basic major diameter of ⅝ in., 20 threads per in., internal member or screw to push, National Buttress form of thread, class 1, right hand.

Note 6 In actual use, excessive gaps between groups should be closed.

Example: 1–16 UN–3A

5.2.3.1 *Gaging practices.* The methods illustrated herein for specifying screw threads imply the acceptability of threads based on the gaging practices set forth in Handbook H28. Whenever the acceptability of threads is premised upon gaging practices neither in accordance with, nor covered by, Handbook H28, the desired basis for acceptability should be specified, either on the drawing providing manufacturing data for the threads, or in a document applicable thereto.

5.2.4 *Methods of designating threads.*

5.2.4.1 *Standard series with standard lengths of engagement.* The limits of size applicable to threads in this category are tabulated in Handbook H28. Accordingly, only the basic designation is required for identification.

EXAMPLES:

External Thread

⅝–16 UNF–2A or .7500–16 UNF–2A

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Internal Thread
¾–16 UNF–2B or .7500–16 UNF–2B

5.2.4.2 *Standard series with non-standard lengths of engagement.* The symbol "UNS" shall apply to threads in this category. Such threads shall be specified using the basic designation, supplemented by the limits of size for the special pitch diameter and the non-standard length of engagement.

EXAMPLE:
¾–16 UNS–2A
PITCH DIA .7079–.7026
LENGTH OF ENGAGEMENT = 1.50–IN. MIN

5.2.4.3 *Special series with standard lengths of engagement.* The limits of size applicable to threads in this category are not tabulated in Handbook H28, but must be computed from the special thread data provided therein. The symbol "UNS" shall apply. Such threads shall be specified using the basic designation, supplemented by the limits of size for the major, pitch, and minor diameters.

EXAMPLES:
External Thread
¾–18 UNS–2A
MAJOR DIA .7486–.7399
PITCH DIA .7125–.7079
MINOR DIA .6804 MAX
Internal Thread
¾–18 UNS–2B
MINOR DIA .690–.708
PITCH DIA .7139–.7199
MAJOR DIA .7500 MIN

5.2.4.4 *Special series with non-standard lengths of engagement.* Threads in this category shall be specified as prescribed by par. 5.2.4.3, except that the non-standard length of engagement shall be included.

EXAMPLE:
External Thread
¾–18 UNS–2A
MAJOR DIA .7486–.7399
PITCH DIA .7125–.7074
MINOR DIA .6804 MAX
LENGTH OF ENGAGEMENT = 1.50–IN. MIN

5.2.4.5 *Stanaard series — modified.* Standard series threads, when modified as set forth in par. 3.8, shall be specified using the basic designation, supplemented by the diameter so modified, each flagged by the abbreviation "MOD"

EXAMPLES:
External Thread
¾–16 UNF–2A–MOD
MAJOR DIA. .7385–.7300–MOD
Internal Thread
¾–16 UNF–2B–MOD
MINOR DIA .686–696–MOD

5.2.4.6 *Special series — modified.* Special series threads, when modified as set forth in par. 3.8, shall be specified using the basic designation, supplemented by the limits of size for the major, pitch, and minor diameters. The abbreviation "MOD" shall be entered following the thread class and the limits so modified.

EXAMPLES:
External Thread
¾–18 UNS–2A–MOD
MAJOR DIA .7380–.7299–MOD
PITCH DIA .7125–.7079
MINOR DIA .6804 MAX
Internal Thread
¾–18 UNS–2B–MOD
MINOR DIA .7005–.7130–MOD
PITCH DIA .7139–.7199
MAJOR DIA .7500 MIN

5.2.4.7 *Standard series or special series — modified and having non-standard lengths of engagement.* The symbol "UNS" shall apply to a standard series or a special series thread, modified as set forth in par. 3.8 and altered due to a non-standard length of engagement. Such threads shall be specified, using the basic designation, supplemented by the dimensions for the major, pitch, and minor diameters, and the length of engagement. The abbreviation "MOD" shall flag the limits that are modified.

EXAMPLES:
Standard Series
¾–16 UNS 2A
MAJOR DIA .7385–.7300–MOD
PITCH DIA .7079–.7026
MINOR DIA .6718 MAX
LENGTH OF ENGAGEMENT = 1.50–IN. MIN

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Special Series
¾–18 UNS–2A
MAJOR DIA .7380 7299–MOD
PITCH DIA .7125–.7074
MINOR DIA .6804 MAX
LENGTH OF ENGAGEMENT = 1.50–IN. MIN

5.2.4.8 *Unified miniature screw thread designation.* The dimensional data for these threads are tabulated in Handbook H28 (1957) Section V. The symbol "UNM" shall apply. Such threads shall be specified using the basic designation, consisting of the nominal size expressed in hundredths of a millimeter and parenthetically the decimal inch equivalent followed by the thread series symbol.

EXAMPLE:
.80 MM (.0315–IN.) UNM

Note:
When a note on the drawing states that all dimensions thereon are expressed in metric units, optionally, the abbreviation "MM" and the decimal inch equivalent may be omitted from the basic designation.

5.2.4.9 *Special threads.*

(a) Special threads for which the limits of size conform in all detail to the Unified formulations or tabulations as prescribed by Handbook H28, are designated "UNS" and shall be identified by the basic designation, supplemented by the limits of size for major, pitch, and minor diameters, and length of engagement, when non-standard.

EXAMPLE:
External Thread
¾–36 UNS–2A
MAJOR DIA .7490–.7485
PITCH DIA .7310–.7271
MINOR DIA .7149 MAX

"Special" category illustrated above arises from the special diameter-pitch combination.

(b) Whenever a standard series, a special series, or a special thread conforming to Unified tabulations or formulations is altered in any detail except as permitted for modified threads, the thread series and thread class symbols shall be omitted from the designation, and the thread shall be specified in the following manner:

EXAMPLE:
¾–36 UNIFIED FORM–SPL
MAJOR DIA .7380–.7350–SPL
PITCH DIA .7284–.7245–SPL
MINOR DIA .7149 MAX
LENGTH OF ENGAGEMENT = .75–IN. MIN

"UNIFIED FORM-SPL" arises from the fact that the limits of size for both the major and pitch diameters deviate from Unified formulations.

(c) Whenever a standard series, or a special series, or a special thread is altered in form to such an extent that it no longer conforms to the Unified tabulations or formulations prescribed by Handbook H28, both the symbol and class shall be omitted from the designation and the thread shall be specified in the following manner:

EXAMPLE:
¾–10 SPL FORM
THREAD ANGLE 55°
MAJOR DIA .7500–.7428
PITCH DIA .6860–.6820
MINOR DIA .6220 MAX
LENGTH OF ENGAGEMENT = .80-IN. MIN

"SPL FORM" arises from the fact that the included angle and the limits of size for the major and pitch diameters deviate from Unified and American National formulations.

5.2.4.10 *Threads of American National form.* Standard threads of American National form, when tabulated in Handbook H28, shall be specified by the method illustrated in Table IV, example 2. All other American National threads for which dimensional limits are not tabulated in Handbook H28 shall be designated in a manner similar to the following example:

External Thread
8¼–16 NS–1
MAJOR DIA 8.2482–8.2356
PITCH DIA 8.2076–8.1981
MINOR DIA 8.1715 MAX
LENGTH OF ENGAGEMENT = 1.50–IN. MIN

5.2.4.11 *Coated threads.* Coating as defined herein, shall be interpreted to mean an addi-

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tive protective metallic finish. Methods of specifying coated threads necessarily vary with the intended application as described in Handbook H28. In general, for Class 2A threads, the allowance provides adequate undercut for coating build-up, whereas for Class 3A threads, the dimensional limits must be reduced to accommodate the coating. Conversely, for some Class 2A thread applications, in order to meet functional requirements, the full allowance must be maintained after coating. Accordingly, whenever necessary, the thread designation for coated threads shall include the pitch diameter limits that will maintain the desired allowance, or will prevent excessive undercut of thread or build-up of coating. Examples of designations for coated threads follow:

(a) *For Class 2A — Coated*

EXAMPLES:

For General Purpose Applications:

¼–20 UNC–2A–COATED

PITCH DIA .2175 MAX AFTER COATING

For critical application where uncoated thread and coating build-up must be controlled:

¼–20 UNC–2A

PITCH DIA { .2164–.2127 (BEFORE COATING) / .2175 MAX (AFTER COATING) }

(b) *For Class 3A — Coated*

EXAMPLES:

For General Purpose Applications:

¼–28 UNF–3A–COATED

PITCH DIA .2268 MAX AFTER COATING

For critical applications where undercut and coating build-up must be controlled:

¼–28 UNF–3A

PITCH DIA { .2258–.2233 (BEFORE COATING) / .2268 MAX (AFTER COATING) }

5.2.4.12 *Specifying gaging practice.* Consistent with par. 5.2.3.1, whenever it is desired to specify on the drawing a gaging practice which differs from that prescribed by Handbook H28, such practice may be indicated as follows:

EXAMPLE:

¼–28 UNF–2A
SINGLE ELEMENT GAGING

5.2.4.13 *Thread inserts.* An example of a drawing designation for a threaded hole to receive a thread insert from Military Standards (example MS 122083) with a ⅜–16 UNC–3B threaded hole, and a one diameter length of insert with tang removed, is as follows:

.390–.398 DIA HOLE, .72–.75 DEEP
TAP .440 MIN DEEP FOR ⅜–16 UNC–3B
HELICAL COIL INSERT PER MS 33537

5.2.5 *Placement of screw thread designations on a drawing.* A screw thread is designated on a drawing by a note with leader and arrow pointing to the thread. If the leader and arrow point to an end view of a thread, the arrow should touch the thread crest line and point to the center of the circle. An example of placement of thread designations on a drawing is illustrated in Figure 9.

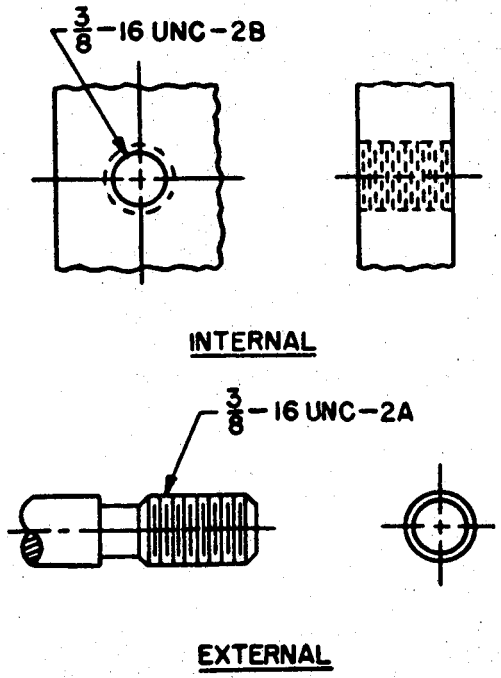


FIGURE 9. *Placement of thread designation*

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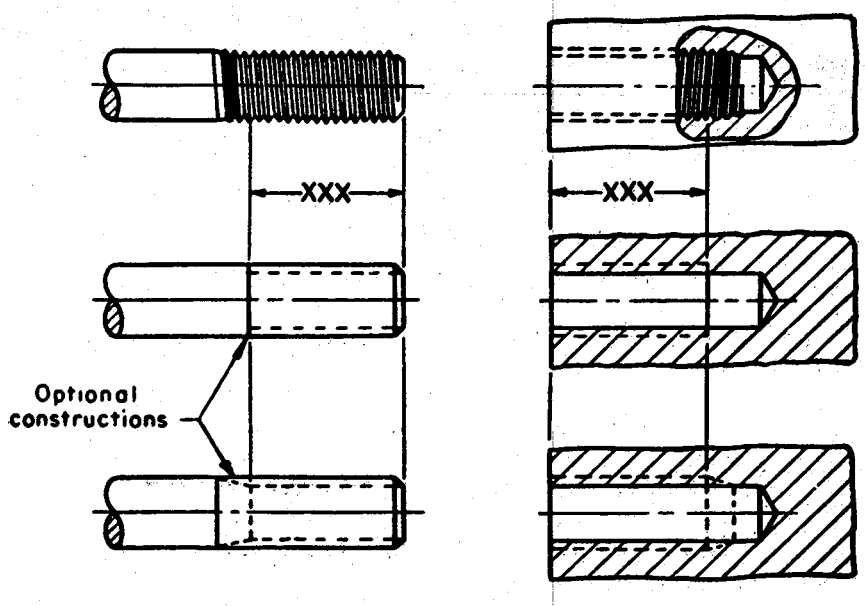


FIGURE 10. *Representation and dimensioning for full form threads with runout not controlled*

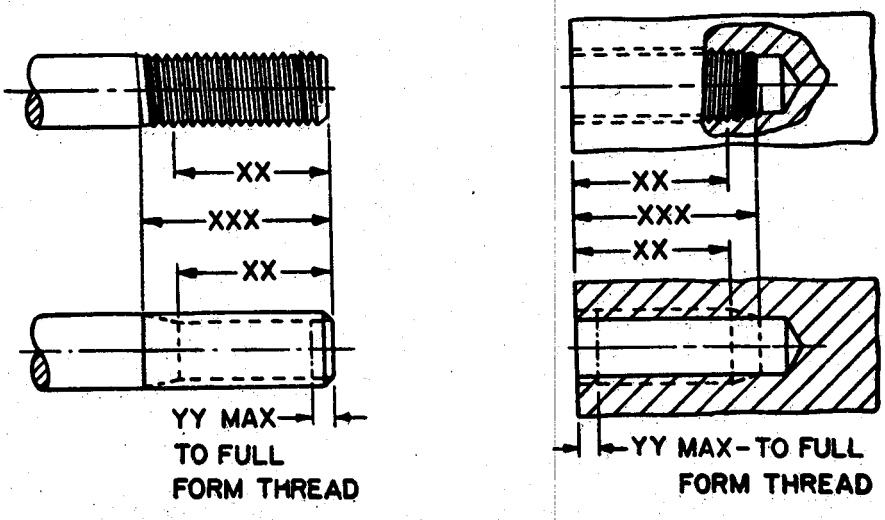


FIGURE 11. *Representation and dimensioning for full form threads with runout controlled*

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5.3 Thread dimensioning.

5.3.1 *General.* For standard series threads, identified as described in paragraph 5.2.4.1, the dimensional limits are obtained by reference to Handbook H28, except those pertaining to length of engagement, chamfers, reliefs, and tap drill diameter and depth. Special series threads and special threads require complete dimensioning in accordance with this standard.

5.3.2 *Full form threads.* The length to be occupied by fully formed threads shall be dimensioned. *When only one dimension is used to specify the length of threads, it shall be interpreted to mean that length occupied by fully formed threads, excluding runout, unless otherwise specified.* Where a chamfer, not exceeding 2 xp in length, exists at the entering end of the thread, it shall be included in the length of fully formed threads, unless otherwise specified.

5.3.3 *Incompletely formed threads.* The length occupied by incompletely formed threads shall not be dimensioned unless essential to design requirements. When not dimensioned this length may vary according to the method of manufacture, consistent with good workmanship.

5.3.3.1 Figure 10 illustrates optional methods of delineating fully formed threads with no limitation on runout.

5.3.3.2 Figure 11 illustrates methods of delineation when both the lengths occupied by fully formed threads and incompletely formed threads must be controlled. Whenever the number of incompletely formed threads allowed on the entering end is more restrictive than permitted by Handbook H28, dimensions imposing such restrictions shall be specified as illustrated in Figure 11, i.e., "YY MAX — TO FULL FORM THREAD.."

5.3.3.3 Figure 12 illustrates an alternative method of delineation when both the length of full form threads and runout must be controlled.

5.3.3.4 *Control of runout.*

5.3.3.4.1 The runout of milled or ground external threads is very abrupt, as shown in Figure 13. The partial threads on the entering end are only a fraction of one thread in length, and the fully formed threads can be run to within one thread length from a shoulder. However, milled and ground threads are costly as compared to die cut or rolled threads.

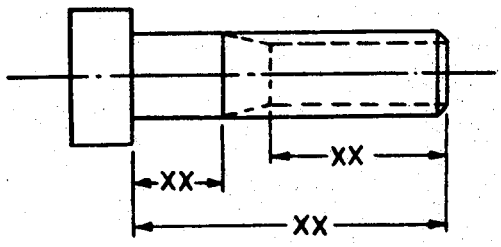


FIGURE 12. *Alternative method of representation when runout must be controlled*

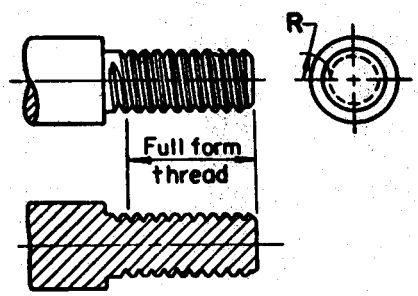


FIGURE 13. *Milled or ground thread*

5.3.3.4.2 Thread cutting dies are usually chamfered two or more threads, producing an equivalent runout of partial threads on the entering end of the product, as shown in Figure 14. The runout on the product can be reduced to 1½ threads by using a die with shorter chamfer or by a second operation with a chaser die having a small chamfer.

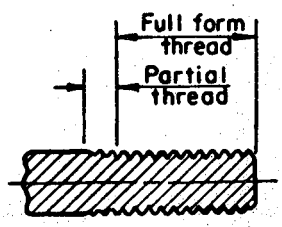


FIGURE 14. *Die cut thread*

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5.3.3.4.3 Thread rolling dies have a minimum chamfer or lead which seldom exceeds one thread in length, producing an equivalent runout on the product, as shown in Figure 15. On unthreaded blanks, where the body diameter approximates the pitch diameter of the thread and where the thread does not abut a head or shoulder, it is generally practicable to limit the runout to one partial thread. If the threads abut a shoulder or head, the distance or allowance from the head or shoulder to the first full form thread should be equivalent to the length of two threads.

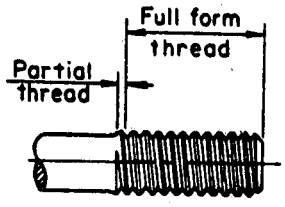


FIGURE 15. *Rolled thread*

5.3.3.4.4 On unthreaded blanks, where the body diameter approximates the major or outside diameter of the thread and where the thread is rolled on a reduced portion of the blank equivalent in diameter to the pitch diameter of the thread, the runout may be limited to 1½ partial threads, providing the threads do not abut a head or shoulder. If the threads abut a head or shoulder, a clearance equivalent to the length of one thread is necessary to avoid kissing the die, and the total distance or allowance to the first full form thread should be equivalent to the length of 2½ threads.

5.3.3.4.5 When information is not available as to the method of manufacture, it is advisable in order not to limit the manufacturing process to provide an allowance between thread and the shoulder or unthreaded shank, not less than that shown in Table VII.

5.3.3.4.6 In the case of blind holes, the allowance for tool chamfer, partial threads and tap clearance at the bottom of the holes should not be less than that shown in Table VII. Where hole depth is restricted by design it should be remembered that in order to produce internal threads of full form to within one thread of the bottom of a hole, the use of a bottoming tap is necessary, involving in some cases a second operation.

5.3.4 *Chamfers.* When chamfers are required, they shall be designated as shown in Figure 16. Whenever practicable, the chamfer angle shall be 45° ±5°, and the point diameter of chamfer shall equal the nominal diameter of thread minus the relief constant B of Table VII (rounded to two decimal places).

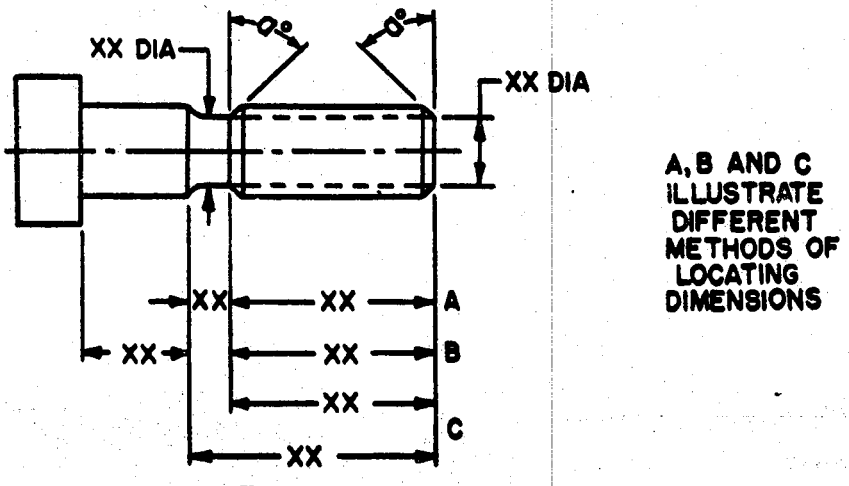


FIGURE 16. *Chamfers and reliefs*

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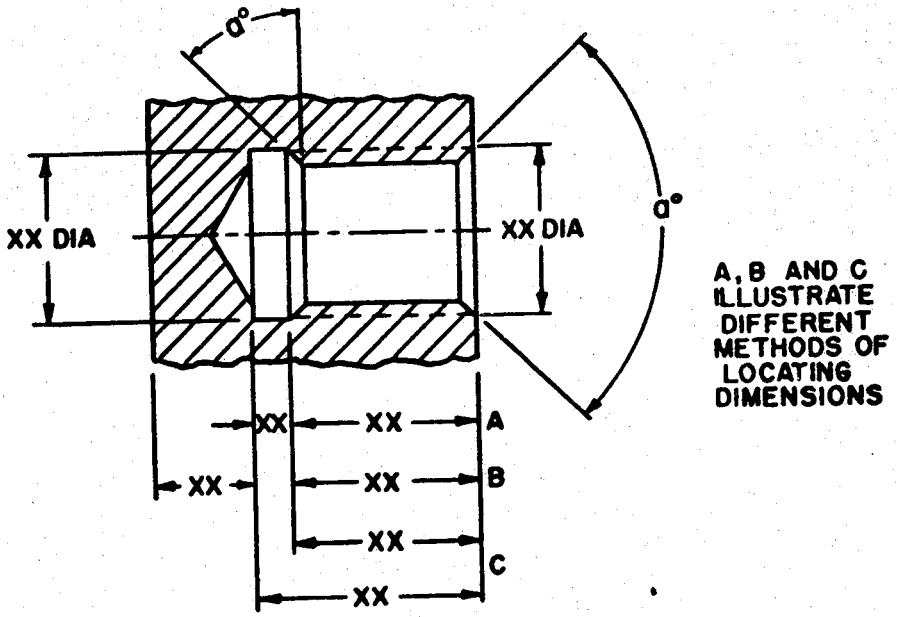


FIGURE 17. *Countersinks and reliefs*

5.3.5 *Countersinks.* Countersinks shall be designated by an angle and a diameter as shown in Figure 17. This angle is normally specified as 90° ±10°. It is recommended that the minimum countersink diameter, as shown in Figure 17, shall be the nominal diameter plus Relief Constant A of Table VII, the final result being rounded to two decimal places.

5.3.6 *Reliefs.* The dimensions for thread reliefs shall be specfiied. External and internal thread reliefs are depicted and dimensioned as shown in Figures 16 and 17. Dimension groups A, B and C show different methods dependent on importance of reference surface. For relief data, see Table VII.

5.3.7 *Tap drill size and depth.* Except when necessitated by specific design considerations, neither the size or depth of the tap drill shall be included in the thread call out. If such data are not specified, both the size and depth of drill may vary at the discretion of the manufacturer provided that the specified limits of size of the thread are not exceeded. Where the depth of the tap drill must be controlled, it may be dimensioned or called out in note form as illustrated in Figures 18 and 19.

5.3.8 *Combined dimensioning.* Complete dimensioning (hole size or tap drill, depth, thread designation, thread depth, countersink, counterbore, etc.) can be accomplished often by a note to a single pictorial presentation. An example of combined dimensioning is shown in Figure 19.

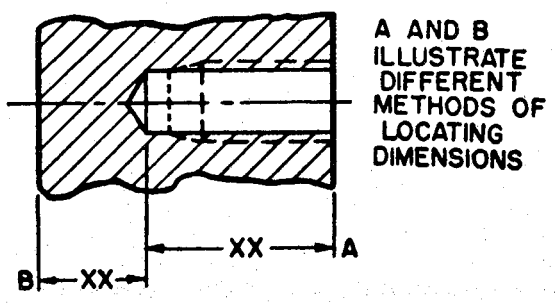


FIGURE 18. *Tap drill depth*

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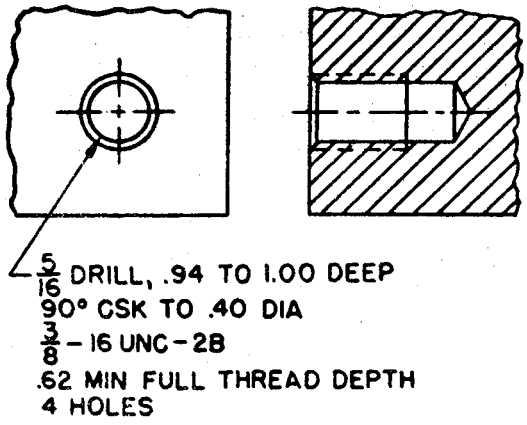


FIGURE 19. *Combined dimensioning*

5.4 Surface finish. Because of the impracticability of determining the roughness height rating of screw threads in production runs, the surface roughness thereof is not normally specified when the surface produced by first class workmanship is considered satisfactory for the purpose intended. However, in those special applications requiring surface control, it shall be specified in accordance with MIL–STD–10.

5.5 Drawing notes. It is recommended that each drawing containing manufacturing data for threads include a reference invoking the documents to be used for interpreting thread dimensions and designations.

EXAMPLE:

"THREAD DIMENSIONS AND DESIGNATIONS SHALL BE INTERPRETED IN ACCORDANCE WITH HANDBOOK H28 AND MIL–STD–9, RESPECTIVELY."

5.6 Recommended selections of threads. Whenever possible, selections of threads should be made from Table I, Standard Series, preference being given to those sizes designated "primary". When the threads listed in Table I do not fulfill design requirements, the next order of preference is the selected diameter-pitch combinations listed in Table II. The third recourse is to compute the limits of size from the table provided in Handbook H28 and the fourth recourse is to apply the formulations provided in Handbook H28. Where design requirements necessitate the use of Unified Miniature screw threads, selection, whenever practicable, from the following list is recommended: .30UNM; .40UNM; .50UNM; .60UNM; .80UNM; 1.00UNM; and 1.20UNM.

5.7 Tables. Tables V, VI and VII are included herein for general information. Table V lists the American Standard thread series for pipe threads, including Dryseal pipe threads, and Table VI lists American National hose coupling and fire-hose coupling threads. Table VII presents suggested dimensional data for screw thread reliefs.

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Copies of this standard may be obtained for other than official use by individuals, firms, and contractors from the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

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Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

Custodians:
Army—Ordnance Corps
Navy—Bureau of Naval Weapons
Air Force

Other interest:
Army—CEMQSigT
Navy—MdShSYMC

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TABLE V. *American standard thread series for pipe threads including Dryseal pipe threads*

	Designation	Nominal size in inches								
		1/16	1/8	1/4	3/8	1/2	.390	3/4	1	1 1/4
		AMERICAN STANDARD PIPE THREADS (EXCEPT DRYSEAL AND HOSE COUPLING TYPES)								
		TPI	TPI	TPI	TPI	TPI	TPI	TPI	TPI	TPI
American Standard taper pipe thread:										
*for general use	NPT	27	27	18		18			14	
for railing joints	NPTR								14	
American Standard straight pipe thread:										
in pipe couplings	NPSC		27	18		18			14	
for loose-fitting mechanical joints with locknuts	NPSL		27	18		18			14	
for free-fitting mechanical joints for fixtures	NPSM		27	18		18			14	
**Special 27 TPI thread of Unified form for use on thin wall tubing	27 UNS			27		27	27	27	27	27
		DRYSEAL AMERICAN STANDARD PIPE THREADS								
Dryseal American Standard:										
fine taper pipe thread	DRYSEAL F-PTF			27	27	27			18	
fuel internal straight pipe thread	DRYSEAL NPSF	27	27	18		18			14	
intermediate internal straight pipe thread	DRYSEAL NPSI	27	27	18		18			14	
taper pipe thread	DRYSEAL NPTF	27	27	18		18			14	
Dryseal SAE short taper pipe thread	DRYSEAL PTF-SAE SHORT	27	27	18		18			14	
Dryseal American Standard:										
special short pipe thread	DRYSEAL PTF-SPL SHORT	27	27	18		18			14	
special extra short pipe thread	DRYSEAL PTF-SPL EXTRA SHORT	27	27	18		18			14	
special diameter-pitch combination pipe thread	DRYSEAL SPL-PTF								27	

* NPT thread also comes in sizes 14, 16, 18, 20, and 24, 8 threads per inch.
** Not a pipe thread. Shown under pipe threads since it is included in the pipe thread section of Handbook H28 — Part II.

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TABLE V. *American standard thread series for pipe threads including Dryseal pipe threads—Continued*

	Designation	Nominal size in inches								
		⅛	¼	⅜	1	1¼, 1½, 2	2½, 3	3½, 4	5, 6	8, 10, 12
		AMERICAN STANDARD PIPE THREADS (EXCEPT DRYSEAL AND HOSE COUPLING TYPES)								
American Standard taper pipe thread:		TPI	TPI	TPI	TPI	TPI	TPI	TPI	TPI	TPI
*for general use	NPT		14		11½	11½	8	8	8	8
for railing joints	NPTR		14		11½	11½	8	8		
American Standard straight pipe thread:										
in pipe couplings	NPSC		14		11½	11½	8	8		
for loose-fitting mechanical joints with locknuts	NPSL		14		11½	11½	8	8	8	8
for free-fitting mechanical joints for fixtures	NPSM		14		11½	11½	8	8	8	
**Special 27 TPI thread of Unified form for use on thin wall tubing	27 UNS	27	27	27	27					
		DRYSEAL AMERICAN STANDARD PIPE THREADS								
Dryseal American Standard:										
fine taper pipe thread	DRYSEAL F–PTF		18		14	14				
fuel internal straight pipe thread	DRYSEAL NPSF		14		11½					
intermediate internal straight pipe thread	DRYSEAL NPSI		14		11½					
taper pipe thread	DRYSEAL NPTF		14		11½	11½	8			
Dryseal SAE short taper pipe thread	DRYSEAL PTF–SAE SHORT		14		11½	11½	8			
Dryseal American Standard:										
special short pipe thread	DRYSEAL PTF–SPL SHORT		14		11½	11½	8			
special extra short pipe thread	DRYSEAL PTF–SPL EXTRA SHORT		14		11½	11½	8			
special diameter-pitch combination pipe thread	DRYSEAL SPL–PTF	27	27	27	27					

* NPT thread also comes in sizes 14, 16, 18, 20, and 24, 8 threads per inch.
** Not a pipe thread. Shown under pipe threads since it is included in the pipe thread section of Handbook H28 — Part II.

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TABLE VI. *American national hose coupling and fire hose coupling threads*

	Designation	Nominal size in inches						
		½	¾	¾	1	1¼	1½	2
		TPI	TPI	TPI	TPI	TPI	TPI	TPI
Garden Hose	NH	11½	11½	11½				
Chemical engine and booster hose	NH			8	8			
Fire Hose	NH						9	
	*NH (SPL)							
Steam, air, water, and all other hose connections to be made up with standard pipe threads	NPSH	14		14	11½	11½	11½	11½
**Special hose coupling threads based on American National pipe threads, NPT	.....							

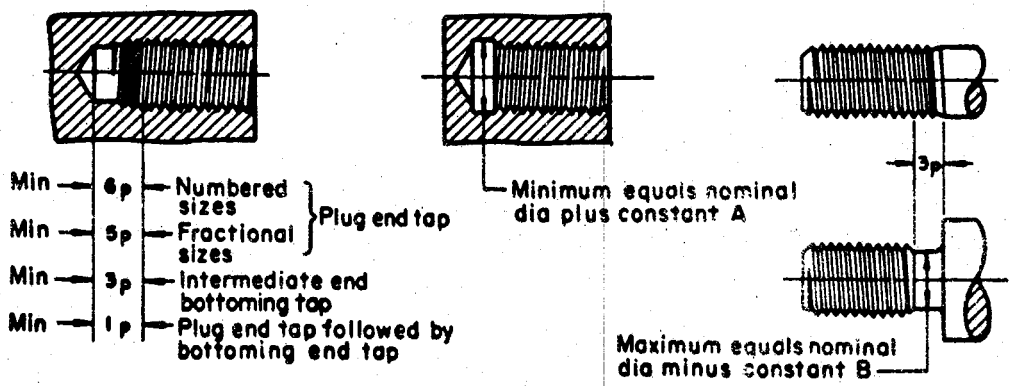
	Designation	Nominal size in inches						
		2½	3	3½	4	4½	5	6
		TPI	TPI	TPI	TPI	TPI	TPI	TPI
Garden Hose	NH							
Chemical engine and booster hose	NH							
Fire Hose	NH	7½	6	6	4	4	4	4
	*NH (SPL)				6			
Steam, air, water, and all other hose connections to be made up with standard pipe threads	NPSH							
**Special hose coupling threads based on American National pipe threads, NPT	.....	8	8	8	8			

* This thread is used extensively by the Navy Department.
** Data on these threads are given in Federal Specification ZZ-H-466, Hose; Gasoline, Rubber-Metal.

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TABLE VII. *Suggested screw thread reliefs*

Threads per inch	Length of relief or allowance beyond full form threads — Pitch Interval							Relief constant	
	1p	$\frac{3p}{2}$	2p	$\frac{5p}{2}$	3p	5p	6p	Internal A	External B
80	.012	.019	.025	.031	.038	.062	.075	.0052	.0196
72	.014	.021	.028	.035	.042	.069	.083	.0056	.0216
64	.016	.023	.031	.039	.047	.078	.094	.0058	.0241
56	.018	.027	.036	.045	.054	.089	.107	.0066	.0272
48	.021	.031	.042	.052	.062	.104	.125	.0074	.0315
44	.023	.034	.045	.057	.068	.114	.136	.0078	.0341
40	.025	.038	.050	.062	.075	.125	.150	.0084	.0373
36	.028	.042	.056	.069	.083	.139	.167	.0090	.0411
32	.031	.047	.062	.078	.094	.156	.188	.0099	.0460
28	.036	.054	.071	.089	.107	.179	.214	.0114	.0526
24	.042	.062	.083	.104	.125	.208	.250	.0126	.0607
20	.050	.075	.100	.125	.150	.250	.300	.0144	.0722
18	.056	.083	.111	.139	.167	.278	.333	.0162	.0814
16	.062	.094	.125	.156	.188	.312	.375	.0180	.0902
14	.071	.107	.143	.179	.214	.357	.429	.0201	.1026
13	.077	.115	.154	.192	.231	.385	.462	.0215	.1103
12	.083	.125	.167	.208	.250	.417	.500	.0232	.1195
11	.091	.136	.182	.227	.273	.455	.545	.0249	.1299
10	.100	.150	.200	.250	.300	.500	.600	.0272	.1427
9	.111	.167	.222	.278	.333	.556	.667	.0300	.1583
8	.125	.188	.250	.312	.375	.625	.750	.0332	.1776
7	.143	.214	.286	.357	.429	.714	.857	.0376	.2026
6	.167	.250	.333	.417	.500	.833	1.000	.0442	.2367
5	.200	.300	.400	.500	.600	1.000	1.200	.0521	.2830
4½	.222	.333	.444	.556	.667	1.111	1.333	.0575	.3141
4	.250	.375	.500	.625	.750	1.250	1.500	.0641	.3528



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